

Causal Inference and Machine Learning

In Economics, Social, and Health Sciences

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From Data to Causality: Alex's Allergy Story

- Alex collects data on meals and symptoms
- Uses **visualizations** to explore relationships
- Forms hypotheses about garlic as the cause

Visual Tools Alex Used

- **Histogram** of reaction intensity
- **Bar Chart**: reactions by food type
- **Time Series**: reactions over time

(Insert example figures here from saved outputs)

Quantifying Correlation

- Alex notices garlic has high correlation with intense reactions
- Some spurious patterns: weekends linked with reactions due to dining out

```
cor(data$garlic, data$reaction_intensity)
```

Controlling for Confounders with Regression

Alex estimates:

```
model <- lm(reaction_intensity ~ garlic + weight + temperature)
summary(model)
```

- Holds constant other variables
- Finds garlic still strongly predicts symptoms

Experimental Design to Identify Causality

- Alex and friends eat same meals (some with garlic)
- Only Alex shows symptoms
- Confirms garlic as causal trigger

Spurious Correlation: Mia's Story

- Mia gets sick on garlic days too
- Turns out: it's the **candle** causing her reactions
- Not garlic → **confounding by context**

From Correlation to Causation

- Step 1: Observe patterns in data
- Step 2: Control for confounders
- Step 3: Test causality experimentally
- Step 4: Rule out alternative explanations

Qualitative vs. Quantitative Research

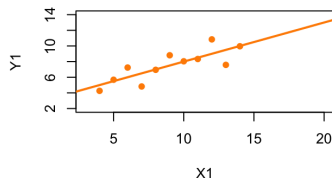
- **Qualitative:** interviews, focus groups, documents
- **Quantitative:** surveys, statistical models, structured data

Data and Visualization

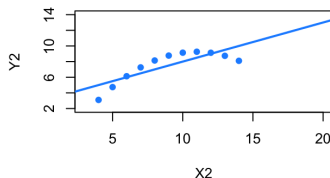
- Visualization helps generate hypotheses
- Patterns can be misleading if not contextualized

Anscombe's Quartet: Why Visualization Matters

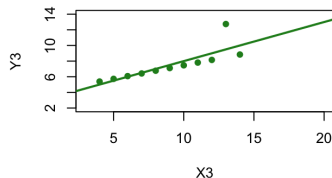
Group I



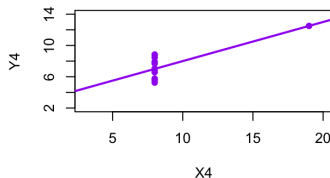
Group II



Group III



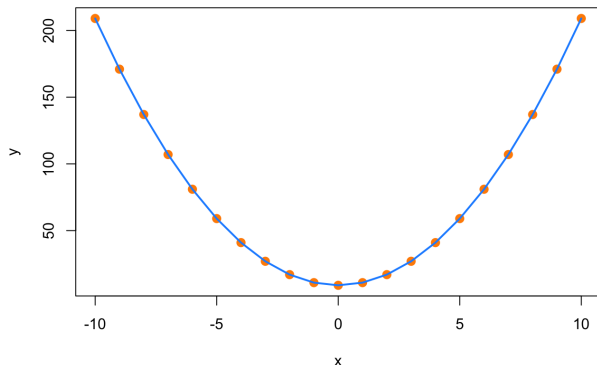
Group IV



- Same mean, variance, correlation, and regression line
- Different visual patterns → Different data stories

Nonlinear Relationships

Non-linear Relationship: $9 + 2 \cdot x^2$



- Correlation may be high, but linear regression fails
- Nonlinearity is a common reason for poor model fit

Correlation and Regression

- Correlation measures association, not direction of effect
- Regression allows prediction and controlling for other variables

The Effect of X on Y

We want to model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

- Y : outcome
- X : predictor
- ε : unobserved error

Estimating β_0 and β_1

```
b1 <- cov(x, y) / var(x)
b0 <- mean(y) - b1 * mean(x)
```

- Use covariance and variance
- Intercept is derived after slope

From Estimation to Prediction

- Estimation: infer relationship in the population
- Prediction: forecast Y for new X

$$\hat{Y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 X_{\text{new}}$$

Causal Effect

- Association $\neq$ Causation
- Need to rule out confounders, reverse causality, selection bias

Example approaches: - Randomized controlled trials - Instrumental variables - Matching, Regression, DiD